

Metrics of Successes: Classification Problem

$$\text{Accuracy} = \frac{\text{correct classifications}}{\text{all classifications}}$$

Macro accuracy=Average the accuracy of all classes

	Positive Prediction	Negative Prediction
Positive Class	True Positive (TP)	False Negative (FN)
Negative Class	False Positive (FP)	True Negative (TN)

Binary classification

$$\text{Accuracy} = \frac{TP+TN}{TP + TN+FP+FN}$$
$$\text{Macro-Averaged Accuracy (MAA)} = 0.5 * \frac{TP}{TP + FN} + 0.5 * \frac{TN}{TN + FP}$$

Used when the dataset is imbalanced as accuracy is biased under such a condition

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	Positive Prediction	Negative Prediction
Positive Class	True Positive (TP)	False Negative (FN)
Negative Class	False Positive (FP)	True Negative (TN)

$$\text{Sensitivity or Recall for the positive class} = \frac{TP}{TP + FN}$$

$$\text{Specificity or Recall for the negative class} = \frac{TN}{FP + TN} \quad \text{Not impacted by class imbalance}$$

$$\text{Precision for the positive class} = \frac{TP}{TP + FP} \quad \text{Precision for the negative class} = \frac{TN}{TN + FN}$$

Precision is a proportion of correct predictions over all predictions

$$F_1 = \frac{2(\text{precision} \times \text{recall})}{\text{precision} + \text{recall}}$$

A higher F1-score would mean that precision and recall are high, while a lower F1-score signifies a high imbalance between precision and recall